



**An open source agentic reconciler framework proven at scale**

Manfred Moser, Sr Principal DevRel Engineer

<https://driftlessaf.ai/>

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The AI age –  
supply chain risk,  
AI-assisted development,  
and the future of DevOps.

# Is DevOps doomed?

富嶽三十六景  
巻第三  
大波瀾  
歌川国芳



# Now add AI to the mix.

More people. More tools. More power. Less context.





- Agentic reconciler framework
- Open source at <https://driftlessaf.ai/>
- Driftless means - no deviation from the desired state
- Agentic framework - AI-powered and traditional bots
- Reconciler like Kubernetes control plane for deployment and YAML-defined state

# Reconciliation examples

Content of GitHub pull requests and content in GitHub repositories

- Ensure new release of a package is used
- Verify PR title and description
- Enforce test coverage and build success
- Improve source code and configuration files

# Why we built DriftlessAF?

- Our event-driven pipeline buckled under the load.
- Every dropped event became a missed package or failed build.
- Engineers were on call for work that should have been automated.
- Open source ships faster every year — we couldn't keep up by hand.
- Security urgency made hours-not-months the new bar.
- We needed reconciliation, not retries.

# Agents: not magic, not marketing – real benefits

- Not chatbots. Not autocomplete. Not “ask it for a script.”
- Workflows with state and goals
- Plan → Act → Verify → Learn → Repeat
- Executing real tasks, in real systems, continuously
- From prompts to pipelines

*That's how reconcilers in DriftlessAF work.*



# Essential lesson: Humans in the loop and accountable

- Systems verify
- Agents propose
- Humans approve
- Trust through visibility
- Accountability isn't removed — it's amplified

# What is included with DriftlessAF

- Complete operating framework
- Terraform recipes for Google Cloud deployment
- Work queue, management tools, observability, dashboards
- Agentic AI framework
- Numerous reconciler bots and examples
- All code written in Go

# Workqueue – architecture notes

- URLs as task key, status data as JSON objects
- Stored in GCS buckets
- Very resilient, per region, optionally sharded
- queued, in-progress and dead-letter states as prefix
- Prioritization and automatic retry support
- Dispatcher to hand work to reconciler bots for concurrent processing

# Agentic AI infrastructure – models and prompts

- Meta-agent layer for Gemini, Claude, and OpenAI
- Executors manage conversation flow with LLM
- Secure prompt builder
  - Parameterized
  - Immutable
  - Support for dynamic binding of JSON, XML, YAML

# Agentic AI infrastructure – tools and evals

- Robust tool-calling mechanism
  - Abstracts model specifics
- Agent evaluation framework for optimal choice and historic tracking including timing, cost, response quality, ...
- Judge – LLM evaluation of LLM output

# Agentic AI infrastructure – advanced use

- Support RAG for improvements based on historical data and engineering input
- Observability with tracing and monitoring tools
  - Including metrics for model interactions



# GitHub reconcilers – operations

- Robust authentication and commit signing
- Supports pull requests and issues
- Find, create, update, close operations
- Embedded, structured metadata from Go templates in description
- Optimized , pooled clone management for git operations

# GitHub reconcilers – file changes

- Supports file path analysis and file modifications
  - Source code
  - Test source code
  - Config files
- Status reporting via GitHub check runs

# Other reconcilers

- Linear issues
- APK Linux packages
- OCI container images

# Example reconcilers

- github-pr-validator - validate title and description
- github-pr-autofix - improve title and description
- github-issue-materializer - create issue
- github-path-modernizer - improve Go code using agent

*It's demo time! Head on over to*

<https://github.com/mosabua/driftlessaf-example>

# Go modernization

## Use `net/http` Status Code Constants

Use named constants from the `net/http` package instead of hardcoded numeric HTTP status codes. Named constants are self-documenting and less error-prone.

✓ Correct:

```
import "net/http"

http.Error(w, "not found", http.StatusNotFound)
w.WriteHeader(http.StatusOK)
if resp.StatusCode == http.StatusUnauthorized {
    // handle unauthorized
}
```

✗ Wrong:

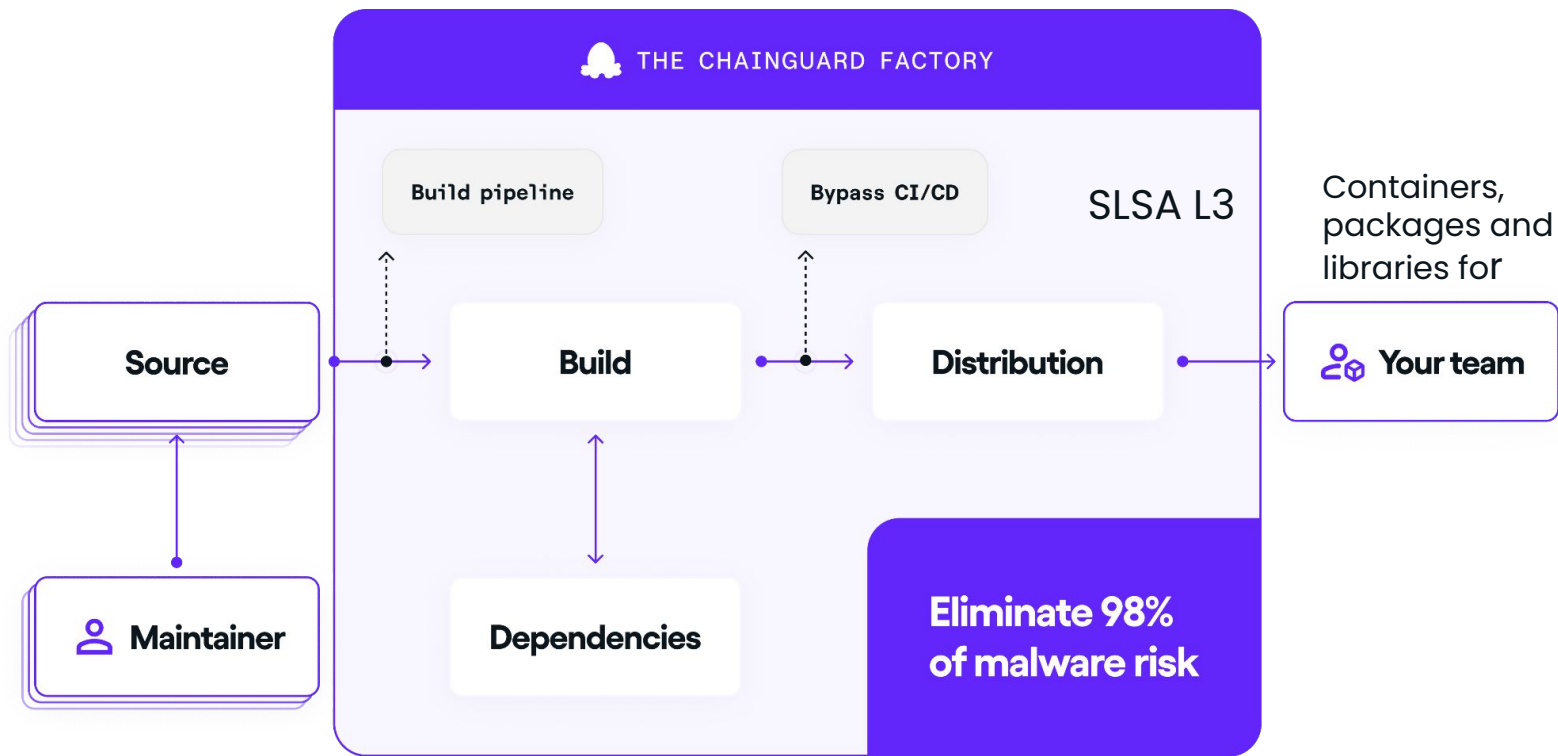
```
http.Error(w, "not found", 404)
w.WriteHeader(200)
if resp.StatusCode == 401 {
    // handle unauthorized
}
```

Audit: Search `.go` files for hardcoded status codes like `\b(200|201|204|301|302|400|401|403|404|405|409|500|502|503)\b` in contexts where `net/http` constants should be used instead.

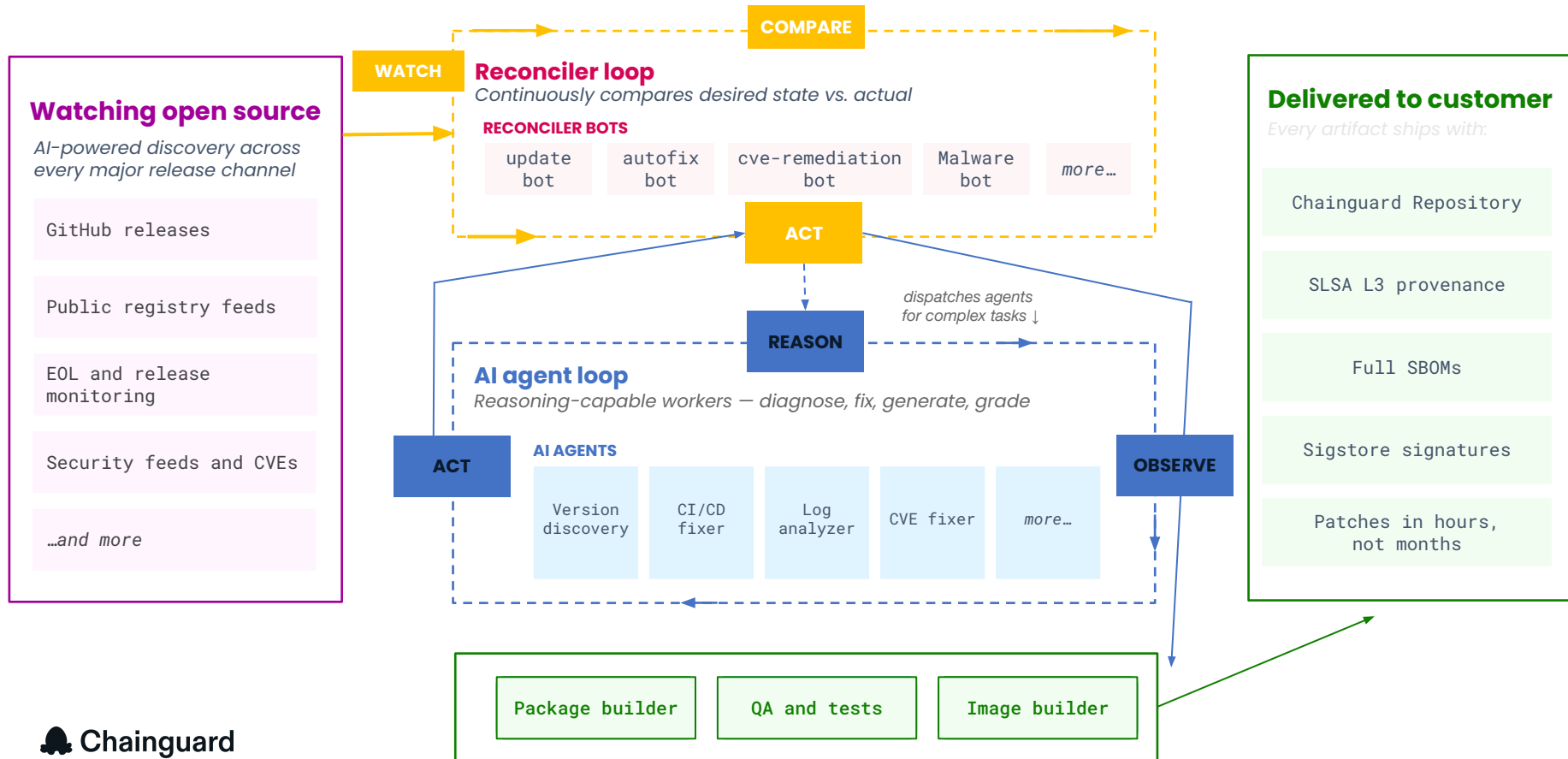


**What we do at Chainguard...**

# Chainguard Agentic Factory

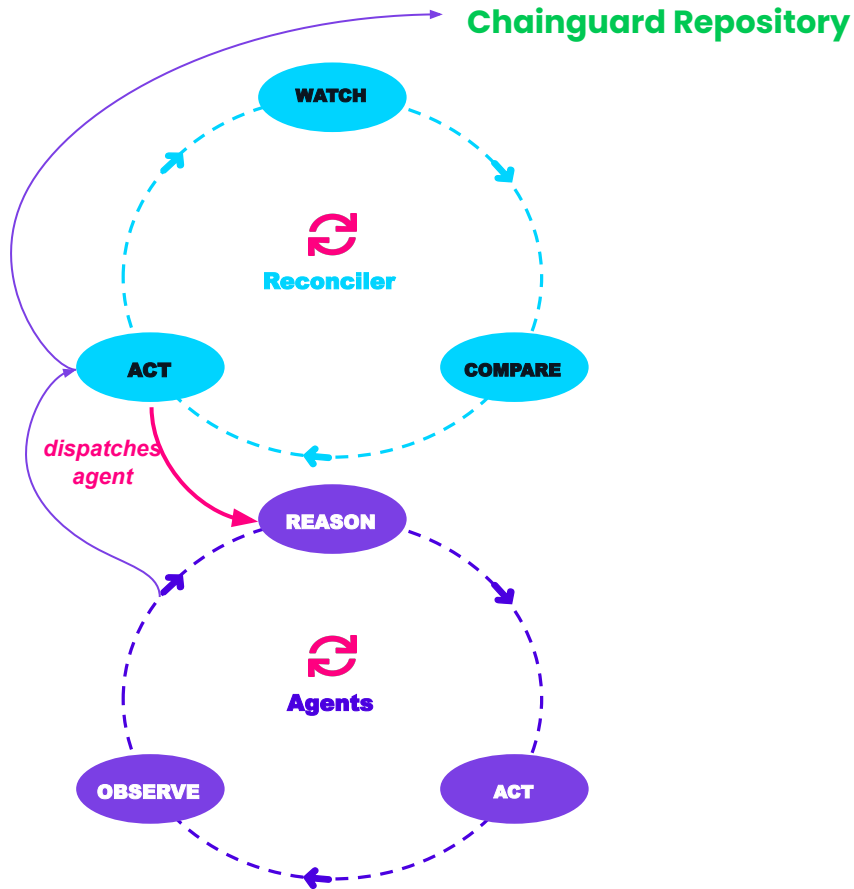


# The Chainguard Factory





# DriftlessAF powering Chainguard Factory



## Reconciler bots

Specialized workers — each watches a specific type of drift

### update bot

Bumps versions

### autofix bot

Patches broken builds

### cve-remediation bot

Patches CVEs

### package bot

Improves tests



## AI agents

Reasoning-capable — diagnose, fix, generate code, grade quality

### Version discovery

Finds new releases

### Fixer

Diagnoses and fixes builds

### Log analyzer

Reads failure logs

### Package improve

Writes better tests

# Scale with DriftlessAF at Chainguard

- Over 2500 container images kept current 24x7
- 238,000+ container versions
- 610M+ container builds
- 1.5 million CVEs remediated
- Thousands of APK packages, npm packages, Python wheels, and Java packages every day

# Wins from DriftlessAF

- Increased stability and throughput
  - Python fixer landed **>2100 PRs at <7500\$**
- Improved test quality and coverage
  - Initial use triggered **900+ PRs** for test improvements
- Faster, more reliable version discovery
  - **From 51% to 100%** coverage
- Build fixes for more libraries and packages from source

# What else could it do?

- Deployments on prem or different cloud providers
- GitLab and others backend systems
- Support for other issue trackers
- Reconcilers for other languages
- More powerful code generation for upgrades
- Reconcile your docs against code and testing

# Projects using DriftlessAF

- Chainguard Factory
- [code-reviewer](#)
- [zen](#)
- Your project next?

# Resources

## Source

- [driftlessaf.ai](https://driftlessaf.ai)
- <https://github.com/driftlessaf/go-driftlessaf>
- <https://github.com/driftlessaf/terraform-infra-reconcilers>

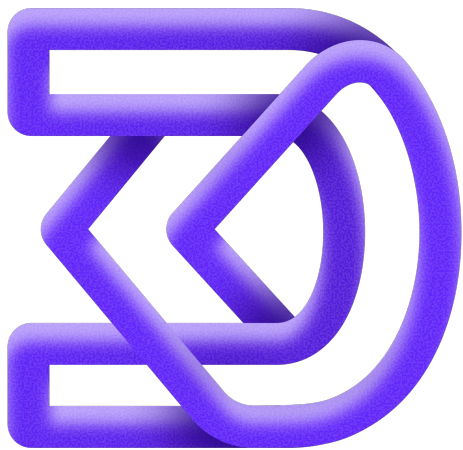
## Additional docs with agent helper:

- <https://codewiki.google/github.com/driftlessaf/go-driftlessaf>
- <https://codewiki.google/github.com/driftlessaf/terraform-infra-reconcilers>

We are hiring for  **DriftlessAF** engineering and beyond



[chainguard.dev/careers](https://chainguard.dev/careers)



# Thank you

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